# Sri Sivasubramaniya Nadar College of Engineering, Chennai

(An Autonomous Institution affiliated to Anna University)

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# Experiment 9

**Clustering Human Activity Recognition Data using K-Means, DBSCAN, and Hierarchical Clustering**

## 1. Aim and Objective

- To implement clustering algorithms on the Human Activity Recognition dataset.
- To apply K-Means clustering and determine the optimal number of clusters using the Elbow Method and Silhouette Score.
- To apply DBSCAN and analyze its ability to detect noise and arbitrary shaped clusters.
- To implement Hierarchical Agglomerative Clustering using Ward's linkage.
- To visualize clusters using dimensionality reduction techniques such as PCA and t-SNE.
- To compare clustering performance using internal and external evaluation metrics.

## 2. Dataset Description

**Human Activity Recognition Using Smartphones Dataset**

- **Source:** UCI Machine Learning Repository
- **Volunteers:** 30 subjects aged between 19 and 48 years
- **Activities:** WALKING, WALKING_UPSTAIRS, WALKING_DOWNSTAIRS, SITTING, STANDING, LAYING
- **Sensors:** 3-axis accelerometer and gyroscope (50 Hz)
- **Window Size:** 2.56 seconds (128 readings per window)
- **Features:** 561 extracted features from time and frequency domains

**Dataset Link:** https://archive.ics.uci.edu

## 3. Data Loading and Preparation

- The dataset was loaded from the UCI HAR Dataset directory.
- Training and testing datasets (`X_train`, `X_test`) were combined into a single dataset.
- Corresponding labels (`y_train`, `y_test`) were merged.
- Feature names were assigned using `features.txt`.
- Activity labels were mapped from numeric values to descriptive names using `activity_labels.txt`.
- The final dataset consists of 10,299 samples and 561 features.

## 4. Preprocessing Steps

- Checked for missing values and confirmed that none were present.
- Standardized all features using `StandardScaler` to ensure uniform scaling.
- Applied dimensionality reduction using PCA and t-SNE for visualization.
- Clustering algorithms were applied on the standardized dataset.

## 5. Exploratory Data Analysis and Visualization

### Feature Distribution:

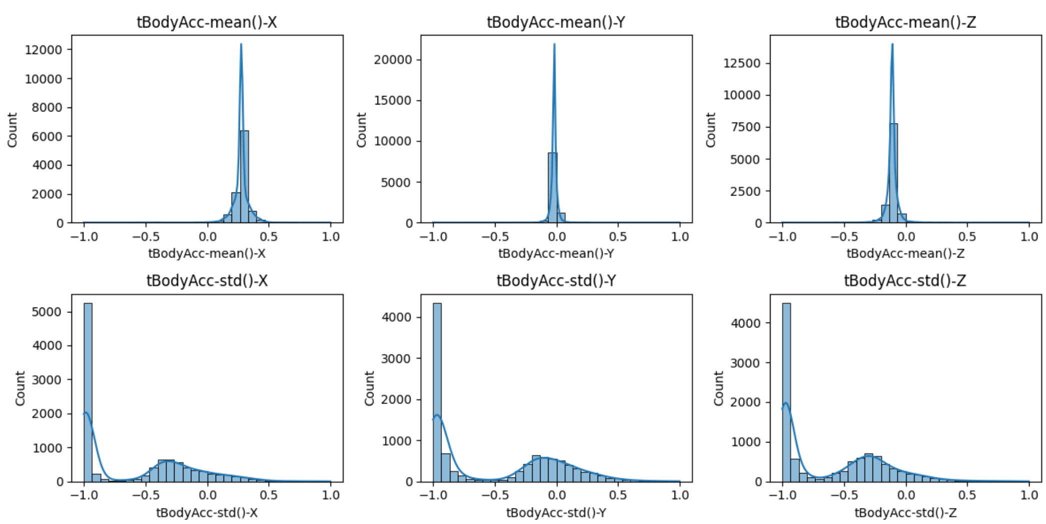
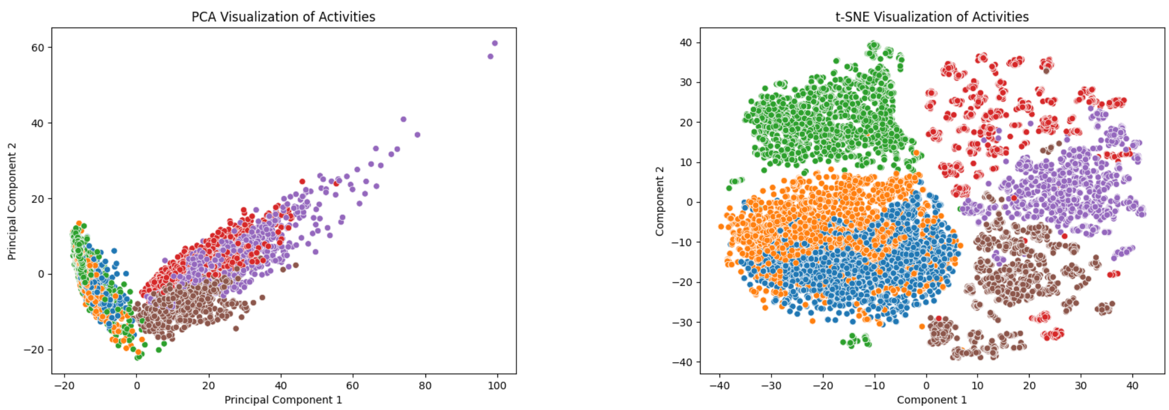



Figure 1: Distribution of Sample Features

**Observation:**
The features show varying distributions, indicating the need for standardization before applying clustering algorithms.

**Dimensionality Reduction using PCA and t-SNE:**




(a) PCA Visualization

(b) t-SNE Visualization

Figure 2: 2D Visualization of Data using PCA and t-SNE

**Observation:**

- PCA shows partial separation of activities.
- t-SNE provides clearer and more distinct clusters.
- Some overlap exists between similar activities like sitting and standing.

## 6. K-Means Clustering

## Elbow Method and Silhouette Analysis

Table 1: K-Means Elbow Method Results

| Number of Clusters ($k$) | WCSS (Inertia) | Silhouette Score |
|---|---|---|
| 2 | 3272856.62 | 0.3937 |
| 3 | 2921077.60 | 0.3155 |
| 4 | 2781602.14 | 0.1500 |
| 5 | 2654789.79 | 0.1286 |
| 6 | 2577180.43 | 0.1099 |
| 7 | 2519774.22 | 0.0816 |
| 8 | 2465378.96 | 0.0749 |

**Observation:**

The WCSS decreases steadily as the number of clusters increases, indicating improved compactness. However, the rate of decrease slows significantly after $k = 3$ or $k = 4$, forming an elbow point. The Silhouette Score is highest at $k = 2$, suggesting well-separated clusters, but this oversimplifies the dataset.

Since the dataset contains six actual activities, choosing $k = 6$ provides better interpretability despite a lower silhouette score.

**Final Choice:** $k = 6$

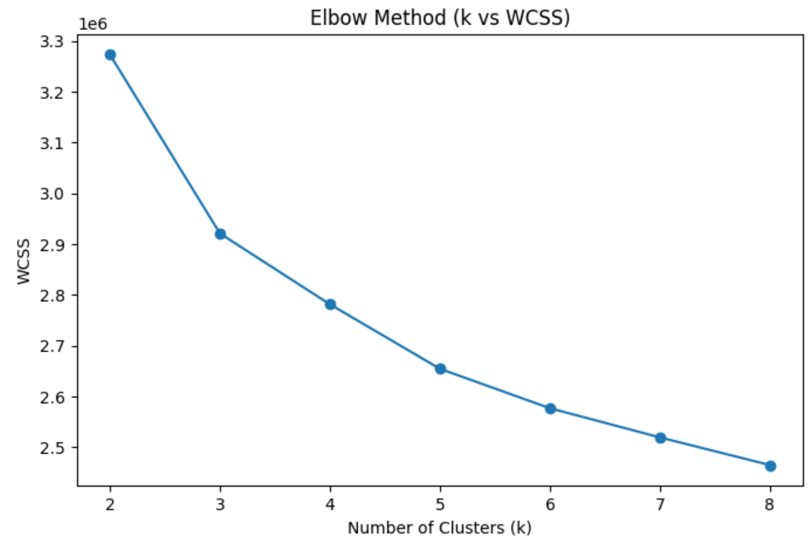



Figure 3: Elbow Curve (k vs WCSS)

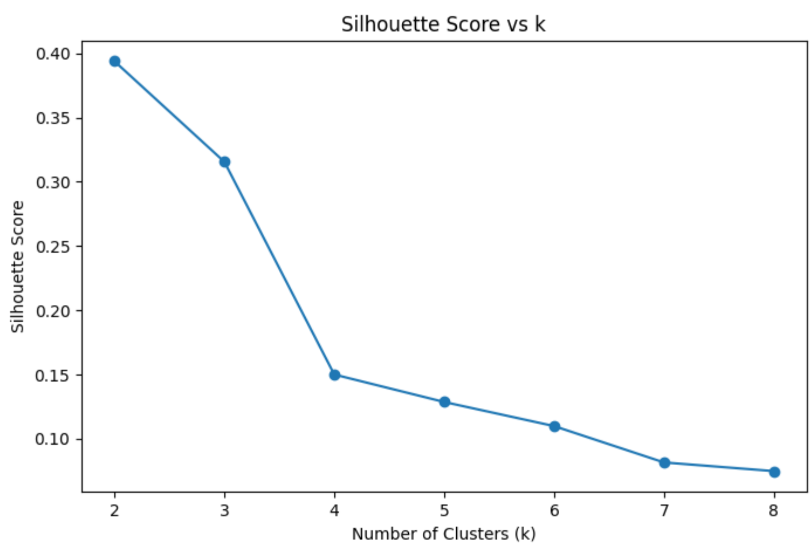



Figure 4: Silhouette Score vs Number of Clusters

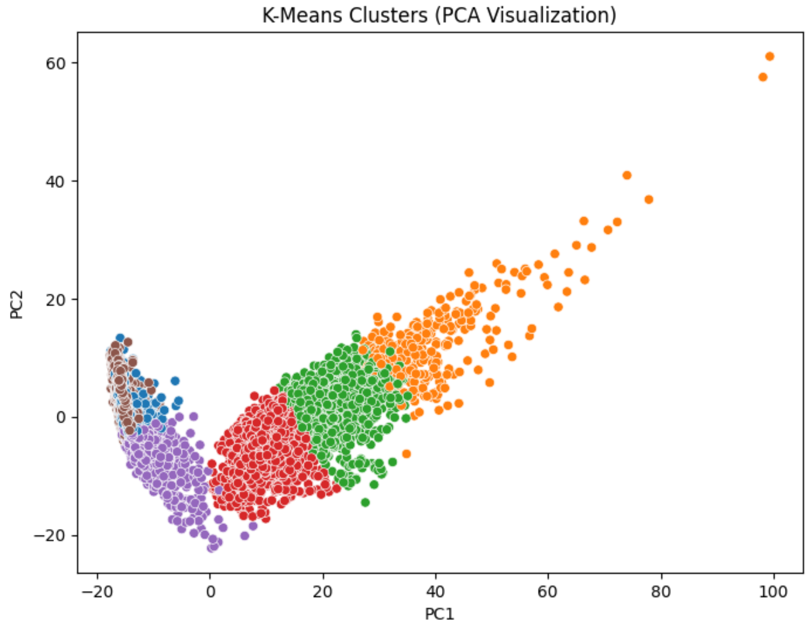



Figure 5: K-Means Clustering Visualization (PCA)

## 7. DBSCAN Clustering

DBSCAN was applied to identify clusters based on density.

**Observations:**

- DBSCAN automatically determines the number of clusters.
- It successfully identifies noise points (labelled as $-1$).
- The algorithm is highly sensitive to the choice of $\varepsilon$ and `minPts`.
- In high-dimensional data, distance measures become less meaningful, leading to poor clustering performance.

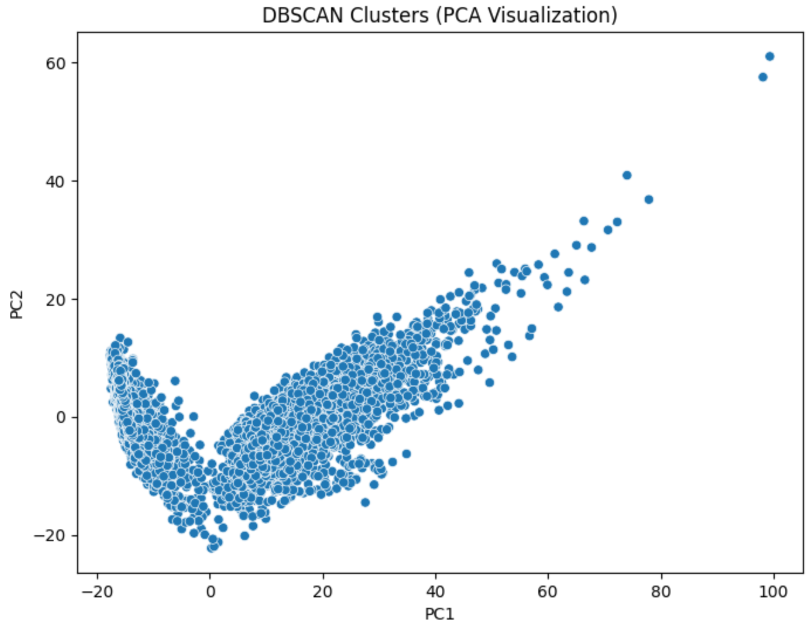



Figure 6: DBSCAN Clustering Visualization (PCA)

**Conclusion:**

DBSCAN struggled to form meaningful clusters for the HAR dataset due to the curse of dimensionality, although it was effective in detecting noise.

## 8. Hierarchical Agglomerative Clustering

Hierarchical clustering was performed using Agglomerative Clustering with Ward's linkage.

**Observations:**

- The dendrogram shows how clusters are merged step by step.
- Ward linkage produces compact and balanced clusters.
- The method does not require random initialization.
- It is computationally expensive compared to K-Means.

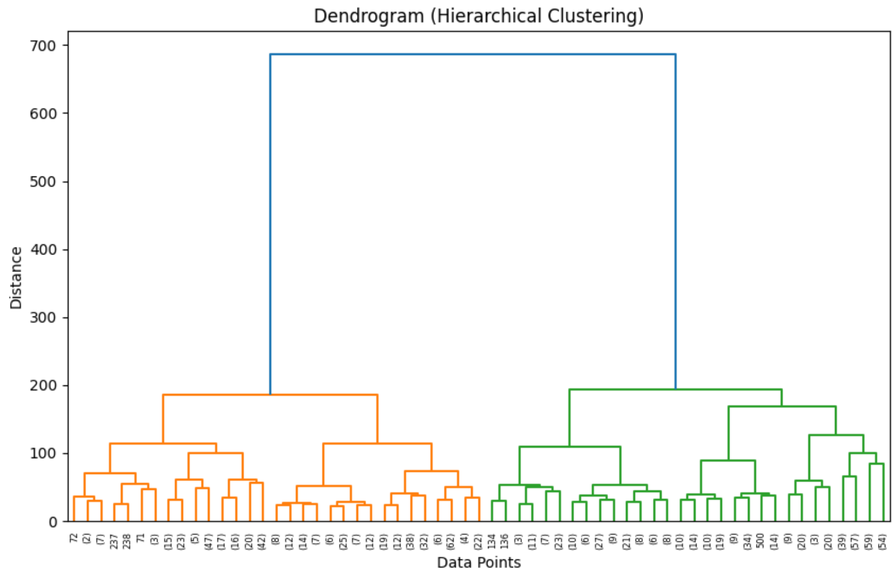



Figure 7: Dendrogram for Hierarchical Clustering

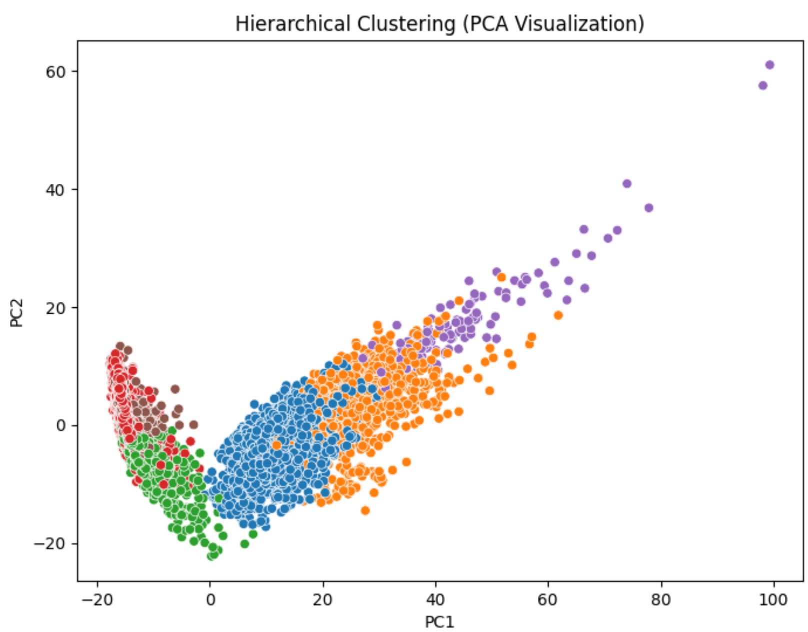



Figure 8: Hierarchical Clustering Visualization (PCA)

**Conclusion:**

Hierarchical clustering provides a clear structure of cluster relationships and performs comparably to K-Means, but at a higher computational cost.

## 9. Evaluation Metrics

The clustering algorithms were evaluated using both internal and external metrics.

**Internal Metrics:**

- Silhouette Score (higher is better)
- Davies–Bouldin Index (lower is better)
- Calinski–Harabasz Index (higher is better)

**External Metrics:**

- Adjusted Rand Index (ARI)
- Normalized Mutual Information (NMI)

**Observation:**

- K-Means achieved the best performance across most metrics.
- Hierarchical clustering showed competitive results.
- DBSCAN performed poorly due to difficulty in parameter tuning and high dimensionality.

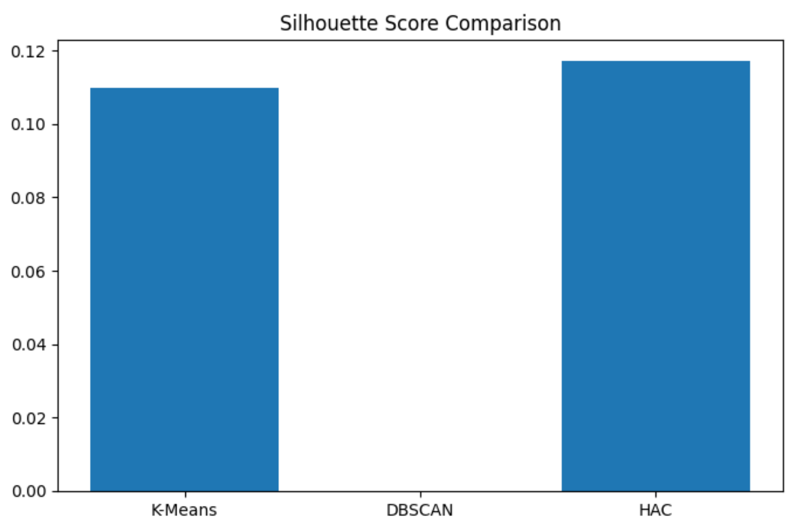



Figure 9: Comparison of Silhouette Scores Across Algorithms

## 10. Observation Questions

**1. Which algorithm produced the most meaningful clusters? Why?**

K-Means produced the most meaningful clusters for this dataset. This is because the HAR dataset, after standardization, forms relatively compact and well-separated groups. K-Means assumes spherical clusters and minimizes intra-cluster variance, which aligns well with the structure of the data. When $k = 6$ is chosen (matching the actual number of activities), the clusters are interpretable and closely correspond to real activity labels.

**2. How sensitive was K-Means to the choice of $k$?**

K-Means was highly sensitive to the choice of $k$. The Elbow Method showed a significant drop in WCSS up to $k = 3$ or $k = 4$, after which improvements slowed down. The Silhouette Score was highest at $k = 2$, but this oversimplified the dataset. Choosing $k = 6$ provided better real-world interpretability. Thus, selecting the correct $k$ is crucial for meaningful clustering.

**3. Did DBSCAN detect noise or small clusters effectively?**

Yes, DBSCAN effectively detected noise points, which were labeled as $-1$. It identified outliers that do not belong to any dense region. However, it struggled to form meaningful clusters due to the high dimensionality of the dataset. Selecting appropriate parameters ($\varepsilon$, minPts) was challenging, leading to either too many noise points or merged clusters.

**4. How does linkage choice affect Hierarchical Clustering?**

The linkage method significantly affects cluster formation:

- **Single linkage:** Can cause chaining effect, leading to elongated clusters.
- **Complete linkage:** Produces compact clusters but is sensitive to outliers.
- **Average linkage:** Provides a balance between single and complete linkage.
- **Ward linkage:** Minimizes variance and produces the most compact and meaningful clusters.

Ward linkage performed best for this dataset.

**5. Which internal metric best matched your visual intuition of cluster quality?**

The Silhouette Score best matched the visual intuition. Higher silhouette values corresponded to well-separated and compact clusters in PCA and t-SNE plots. The Davies–Bouldin Index also reflected cluster quality (lower is better), but was less intuitive. The Calinski–Harabasz Index was useful but harder to interpret directly.

## 11. Final Observations

- K-Means produced the most meaningful clusters due to compact and spherical cluster structure.
- K-Means is sensitive to the choice of $k$, requiring careful tuning.
- DBSCAN effectively detected noise but failed to form clear clusters.
- Ward linkage in hierarchical clustering produced better clusters compared to other linkage methods.

- Silhouette Score aligned well with visual cluster separation but favored fewer clusters.

## 12. Conclusion

Three clustering algorithms – K-Means, DBSCAN, and Hierarchical Clustering – were applied to the Human Activity Recognition dataset.

K-Means performed the best, producing well-separated and meaningful clusters when $k = 6$ was chosen. Hierarchical clustering also performed well and provided additional interpretability through dendrograms.

DBSCAN was less effective due to the high dimensional nature of the dataset, although it successfully identified noise points.

Overall, the choice of clustering algorithm depends on the dataset characteristics, and parameter tuning plays a crucial role in performance.